

HYEONG JIN HYUN

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EMPLOYMENT

Postdoctoral Scholar

Purdue University, 2025 – Present

- Principal Investigators: Dr. Xiao Wang, Dr. Haibo Liu

EDUCATION

Purdue University

U.S.A, 2025

- *Ph.D. in Statistics*
- Advisor: Dr. Xiao Wang

Seoul National University

South Korea, 2020

- *M.S. in Statistics*
- Advisor: Dr. Johan Lim

Seoul National University

South Korea, 2018

- *B.S. in Mathematics and Statistics (double major)*
- Cum Laude
- Leave of absence for military service (2014 - 2016)

RESEARCH INTERESTS

Simulation-based inference, approximate Bayesian computation, conformal prediction, generative model, flow model, machine learning theory, uncertainty characterization

PUBLICATIONS

- **Hyun, H.**, and Wang, X. (2025+). Neural Conformal Inference for Jump Diffusion Processes, *Journal of Econometrics*, accepted, [paper link].
- **Hyun, H.**, and Wang, X. (2025+). Fast Cost-Constrained High Dimensional Regression, *Statistica Sinica*, accepted, [paper link].
- **Hyun, H.***, Kim, Y*., Kim, S., Kim, J., Lim, J., Lim, D. and Kwon, S. (2021). Constrained Principal Component Analysis with Stochastically Ordered Scores for High-Dimensional Mass Spectrometry Data, *Chemometrics and Intelligent Laboratory Systems*, [paper link].

WORKING PAPERS

- (with Wang, X.) Neural Amortized Bayesian Computation (*under review, Journal of the American Statistical Association*).
- (with Liu, Y., Wang, X.) Physics-Informed Neural Networks for Scalable and Accurate ODE Parameter Estimation (*manuscript in preparation*).
- (with Wang, X., Liu, H.) Neural Posterior Estimated Amortized Bayesian Computation (NPE-ABC) (*manuscript in preparation*).

AWARDS

2025 William J. Studden Publication Award	Purdue, April 2025
2024 ICSA Best Poster Award	ICSA, June 2024
2023 ASA Best Student Paper Award B&E Section	ASA, Aug 2023
2023 ICSA Best Student Paper Award Honorable Mention	ICSA, June 2023

TEACHING EXPERIENCE

- Instructor
 - STAT 598: Qualifying exam preparation course Purdue, 2024-2025
Designed and taught a preparatory course in mathematical statistics for first-year Ph.D. students.
Provided detailed solutions to past 15 years of qualifying exams to enhance student readiness.
- TA Coordinator SNU, 2018-2019
Organized and hosted meetings to allocate teaching assistants to classes within the statistics department. Facilitated networking and discussions between TAs and instructors.
- Teaching Assistant
 - STAT 301: Elementary Statistical Methodology Purdue, 2020-2023
Led weekly lab sessions to teach statistical methods and demonstrated SPSS functionality. Designed, distributed, and graded lab assignments to enhance student learning.
 - STAT 519: Introduction to Probability Theory Purdue, 2022
Provided personalized guidance during weekly office hours and proctored exams.
 - STAT 512: Applied Regression Analysis Purdue, 2020
Held lab sessions to learn how to implement regression techniques using R.
 - Statistical Consulting and Practices SNU, 2019
Advised clients on statistical issues related to their research projects.
 - Mathematical Statistics SNU, 2018
Held weekly six-hour recitation, teaching problem-solving strategies for exercises.

TALKS

2024 Purdue GSO	West Lafayette, Nov 2024
2024 JSM Topic-Contributed Session	Portland, Aug 2024
2024 ICSA Poster Session	Nashville, Aug 2024
Purdue GSO Seminar	West Lafayette, Nov 2023
2023 ICSA Symposium Best Student Paper Award Presentation	Ann Arbor, June 2023
2023 JSM Best Student Paper Award Presentation	Toronto, Aug 2023

SKILLS & LANGUAGES

Languages	Korean (native), Jejuo (native), English (fluent)
Computation Skills	Python, R, MATLAB, SPSS
Document Software	LaTeX

SOFTWARE

- NCoin-JDP: <https://github.com/anonymous1116/NCoin-JDP>
- FCR: <https://github.com/anonymous1116/FCR>
- os-PCA: <https://github.com/hjhyu0081/osPCA>